

No. of Functional Features Included in the Solution

Date	25 June 2026
Team ID	XXXXXX
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Functional Features Overview

This document lists all functional features implemented in the Emotion Detection & Learning Support Engine. Each feature contributes to intelligent emotion recognition, AI-assisted learning support, analytics, and an interactive user experience.

S.No	Feature Name	Feature Description	Module / Component	Status	Marks
1	Emotion Detection	Detects student emotions from text using NLP and Machine Learning.	ML Pipeline	Done	✓
2	Text Preprocessing	Cleans and prepares user input before prediction.	Preprocessing Module	Done	✓
3	Generative AI Response	Uses Google Gemini API to generate personalized learning guidance.	Gemini API	Done	✓
4	Learning Recommendation	Provides emotion-aware study suggestions.	Recommendation Engine	Done	✓
5	Interactive Dashboard	Displays emotion predictions and AI responses through Streamlit.	Frontend	Done	✓
6	Emotion Analytics	Shows emotion trends and statistics using charts.	Plotly Dashboard	Done	✓
7	Session History	Stores previous interactions for later review.	Backend	Done	✓
8	Model Evaluation	Evaluates model accuracy and prediction performance.	ML Evaluation	Done	✓
9	Responsive User Interface	Modern Streamlit interface with clean navigation.	Frontend UI	Done	✓

Feature Summary

Metric	Count / Value
Total Features Planned	9
Total Features Implemented	9
Core / Must-Have Features	7
Additional / Nice-to-Have Features	2
Features Tested & Verified	9

Feature Category Breakdown

S.No	Category	Features in Category	Example Features
1	User Interface	3	Dashboard, Input Form, Results Page
2	Machine Learning	2	Emotion Detection, Model Evaluation
3	Generative AI	1	Gemini AI Response Generation
4	Analytics	2	Emotion Charts, Session History
5	Backend Integration	1	Data Processing and API Integration